

JavaScript Learning Notes — Complete So Far

Variables se lekar DOM Todo List ke Edit feature tak.

Variables & Data Types

```
let name = "Nitin";
const age = 25;
let active = true; // boolean
let x;             // undefined
let y = null;
let user = {};     // object
let nums = [];     // array
```

Operators & if/else

```
10 % 3; // 1

if (age >= 18) {
  console.log("Adult");
} else {
  console.log("Minor");
}

// === strict comparison
10 === 10; // true
```

Functions

```
function add(a, b) {
  return a + b;
}

let result = add(10, 20); // 30
```

Arrays & Loops

```
let numbers = [10, 20, 30];

numbers.push(40);

for (let i = 0; i < numbers.length; i++) {
  console.log(numbers[i]);
}

for (let number of numbers) {
  console.log(number);
}
```

Array Methods

```
// forEach = action
numbers.forEach(function(number) {
  console.log(number);
});

// map = transform
let doubled = numbers.map(function(number) {
  return number * 2;
});

// filter = choose
let even = numbers.filter(function(number) {
  return number % 2 === 0;
});

// reduce = combine
let total = numbers.reduce(function(sum, number) {
```

```
    return sum + number;
  }, 0);
```

Chaining

```
let result = numbers
  .filter(function(number) {
    return number % 2 === 0;
  })
  .map(function(number) {
    return number * 2;
  })
  .reduce(function(sum, number) {
    return sum + number;
  }, 0);
```

Objects

```
let user = {
  name: "Nitin",
  age: 25,
  city: "Panchkula"
};

console.log(user.name);
user.age = 26;
user.email = "test@example.com";
delete user.city;
```

Nested Objects & Arrays of Objects

```
let user = {
  name: "Nitin",
  address: {
    city: "Panchkula",
    pin: 134109
  }
};

console.log(user.address.city);

let users = [
  {name: "A", age: 20},
  {name: "B", age: 25}
];

console.log(users[0].name);
```

Object Methods

```
Object.keys(product);
Object.values(product);
Object.entries(product);

Object.entries(product).forEach(function([key, value]) {
  console.log(key, value);
});
```

Destructuring

```
let colors = ["red", "green", "blue"];
let [a, b, c] = colors;

let user = {name: "Nitin", city: "Panchkula"};
let {name, city} = user;

let {name: userName} = user;

function showProduct({name, price, brand}) {
  console.log(name, price, brand);
}
```

```
}
```

Spread & Rest

```
let newFruits = [...fruits, "Banana"];
let updatedUser = {...user, age: 27};

function add(...numbers) {
  return numbers.reduce(function(sum, n) {
    return sum + n;
  }, 0);
}

let [first, second, ...rest] = [10, 20, 30, 40];
```

DOM Selection, Events & Text

```
let title = document.getElementById("title");
let btn = document.querySelector("#btn");

btn.addEventListener("click", function() {
  title.textContent = "Hello Nitin";
});
```

Input & Forms

```
let value = input.value;

if (value.trim() === "") {
  console.log("Please enter value");
} else {
  let num = Number(value);
  console.log(num);
}

form.addEventListener("submit", function(event) {
  event.preventDefault();
});
```

classList

```
box.classList.add("red");
box.classList.remove("red");
box.classList.toggle("red");
box.classList.contains("red");
```

Dynamic CSS with .style

```
box.style.color = "red";
box.style.backgroundColor = "blue";
box.style.padding = "40px";
box.style.borderRadius = "20px";
box.style.fontSize = "30px";
```

Counter Mini Project

```
let count = 0;

plusbtn.addEventListener("click", function() {
  count++;
  value.textContent = count;
});

minusbtn.addEventListener("click", function() {
  count--;
  value.textContent = count;
});

reset.addEventListener("click", function() {
  count = 0;
});
```

```
    value.textContent = count;
  });
```

Dynamic Elements: createElement / append / remove

```
let li = document.createElement("li");
li.textContent = input.value;
list.append(li);

li.remove();
```

Todo: Add + Delete + Complete

```
let li = document.createElement("li");

let span = document.createElement("span");
span.textContent = input.value;

let btn = document.createElement("button");
btn.textContent = "Delete";

btn.addEventListener("click", function() {
  li.remove();
});

span.addEventListener("click", function() {
  span.classList.toggle("line");
});

li.append(span);
li.append(btn);
todo.append(li);
```

Todo Complete CSS

```
.line {
  text-decoration: line-through;
  color: gray;
}
```

Todo Edit: Dynamic Input + Done

```
let edit = document.createElement("button");
edit.textContent = "Edit";

edit.addEventListener("click", function() {
  let editinput = document.createElement("input");
  editinput.type = "text";
  li.append(editinput);

  let done = document.createElement("button");
  done.textContent = "Done";
  li.append(done);

  done.addEventListener("click", function() {
    let newText = editinput.value;

    if (newText.trim() !== "") {
      span.textContent = newText.trim();
    }

    done.remove();
    editinput.remove();
  });
});

li.append(edit);
```

Quick Revision

map = change • filter = choose • reduce = combine • spread = expand • rest = collect

DOM: select → event → createElement → textContent → append → remove

Next: keyboard events — Enter press karke Todo add karna.